MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)
MG3VACMAT200-MASTERING PROBLEM SOLVING THROUGH VEDIC MATHEMATICS

Duration: 75 Minutes Maximum Marks: 50

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Answer any five questions. Each question carries 2 marks. Each question must be solved using the relevant Vedic sutras with proper steps.

1. Add: 486 + 654 + 987 [S][2]
2. Name two vedic methods for subtraction [K][1]
3. Multiply 147 by 99. [A][2]
4. Find the square of 165. [A][5]
5. If 15 men can reap a field in 28 days, in how many days will 5 men reap it?. [An][5]
6. Find the simple interest on 400 for 5 years at 6 per cent. [A][4]
7. Multiply $(8x - 3y)$ by $(2x + 4y)$. [A][7]
8. Solve $(x + 1)(x + 2) = (x - 3)(x - 4)$ [A][3]

Part B
Answer any four questions. Each question carries 5 marks. Each question must be solved using the relevant Vedic sutras with proper steps.

9. Add: 75934 + 87628 + 34879 + 14093 [A][2]
10. Subtract 375988 from 823479. [An][2]
11. Find the square of 687. [A][2,5]
12. Find the square root of 20736. [An][2,5]
13. Multiply $2x^2 - 4x + 6$ by $3x^2 - 7x - 2$. [A][7]
14. Solve $\frac{2}{2x+1} + \frac{18}{3x+1} + \frac{75}{5x+1} = \frac{88}{4x+1}$. [An][3]

Part C

Answer any two questions. Each question carries 10 marks. Each question must be solved using the relevant Vedic sutras with proper steps.

15. (a) Multiply 659 by 898 [An][2]

    (b) Divide 387 by 32 [A][2,5]

16. (a) Find the cube of 58 using Nikhilam Sutra. [A][2]

    (b) Find the cube root of 84604519. [An][2]

17. Determine whether 2774 is divisible by

    (a) 19

    (b) 49 [S][2]

18. Solve the system of linear equations

    (a) $x - y = 7$
    $5x + 16y = 14$. [An][7,8]

    (b) $12x + 8y = 7$
    $16x + 16y = 14$. [An][7,8]